Q-Learning Based Reinforcement Learning for Multi-Agent Foraging: Simulation Report

1. Introduction and Objectives

Reinforcement Learning (RL) is a learning paradigm where an agent learns optimal behavior through interaction with an environment by maximizing cumulative rewards. One of the most widely used RL algorithms is **Q-learning**, which allows agents to learn action-value functions without requiring a model of the environment.

This project explores the behavior of **foraging agents** in a grid-world environment using Q-learning. The study focuses on both **single-agent** and **multi-agent** settings to understand how learning scales when multiple agents interact within the same environment.

Objectives

- To study the learning behavior of a **single agent** using Q-learning.
- To analyze **multi-agent learning** using:
    - Independent Q-tables.
    - A shared Q-table.
- To compare **cooperation vs. competition** in multi-agent systems.
- To evaluate performance using reward trends, food collection, and collision rates.

2. Q-Learning Parameters and Justification

Q-learning updates the Q-value using the following rule:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max\ Q(s',a') - Q(s,a)]$$

Parameters Used

- **Learning rate (α):** Controls how much new information overrides old knowledge.
  A moderate value ensures stable learning without sudden fluctuations.
- **Discount factor (γ):** Determines the importance of future rewards.
  A high value encourages long-term reward optimization.

- **Exploration rate (ε):** Controls the agent's tendency to explore using an ε-greedy policy. ε decays gradually after each episode to shift from exploration to exploitation, but it is bounded by a minimum value (min_epsilon=0.01) to ensure occasional exploration throughout training
- **Number of episodes:** Experiments were conducted using both **200** and **1000** episodes to study convergence behavior.

These parameters were chosen to balance **learning stability**, **exploration**, and **convergence speed**

3. Single-Agent Learning Behavior

In the single-agent setup, one agent navigates a 10×10 grid to collect food while avoiding obstacles.

Observed Behavior

- Initially, the agent explores the environment randomly, resulting in low rewards.
- As learning progresses, the agent discovers optimal paths toward food sources.
- The total reward per episode increases steadily, indicating successful learning.
- With 1000 episodes, the learning curve becomes smoother and more stable compared to 200 episodes
- The episode length (number of steps taken per episode) decreases over time, showing that the agent learns to reach food more efficiently and avoid collisions

Analysis

- The agent learns to minimize unnecessary movements.
- Exploration decreases over time due to ε decay.
- Longer training leads to better convergence and higher average rewards
- Monitoring episode length provides insight into learning efficiency, complementing reward and food collection metrics.

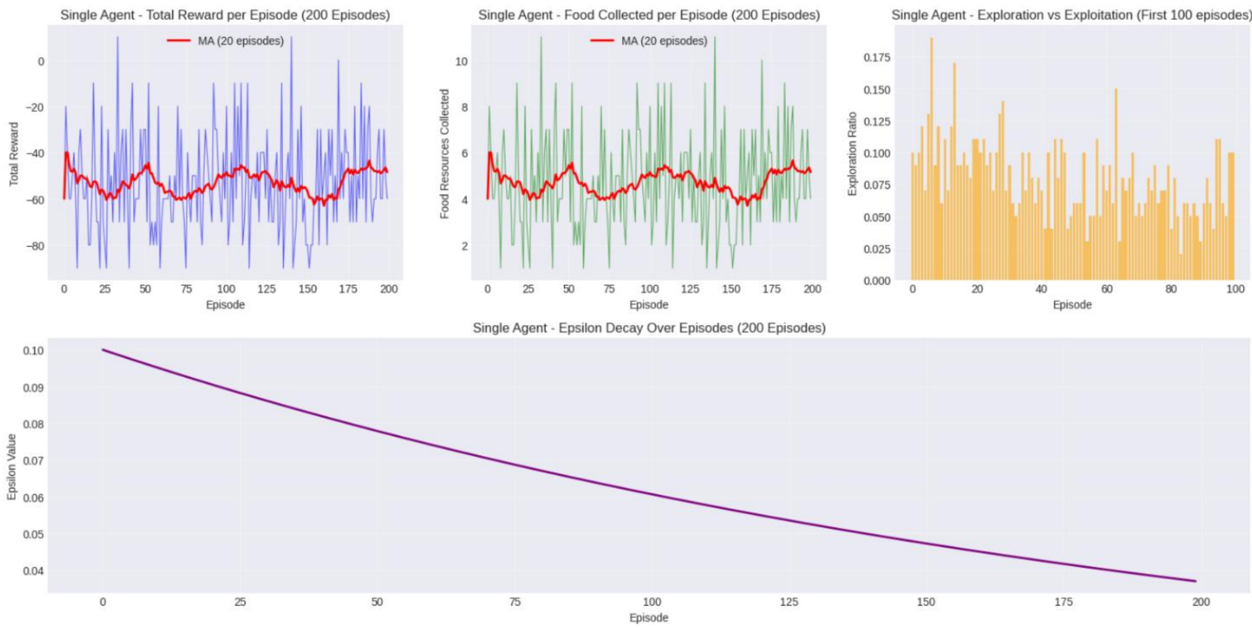

Figure1:Single Agent Learning Performance Analysis (200 Episodes)

The figure contains these four panels:

1. **Top-left**: Total reward per episode with moving average
2. **Top-right**: Food resources collected per episode with moving average
3. **Bottom-left**: Exploration ratio (random vs greedy actions) for first 100 episodes
4. **Bottom-right**: Epsilon decay over 200 episodes

All these plots together provide a comprehensive view of how the single forager agent learned to collect food resources using Q-learning over 200 training episodes

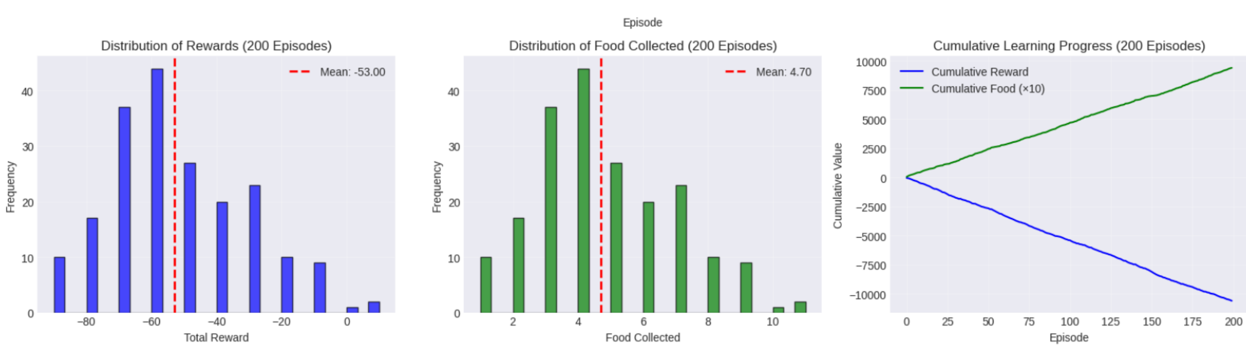

Figure2:Single Agent Performance Distributions and Cumulative Progress (200 Episodes)

The figure contains these three panels:

1. **Left**: Histogram of reward distribution with mean value

2. **Middle**: Histogram of food collected distribution with mean value
3. **Right**: Cumulative learning progress showing both cumulative reward and cumulative food (scaled by 10)

These plots provide a **statistical analysis** of the agent's performance, showing:

- How rewards are distributed across episodes
- How food collection amounts vary
- The overall learning trajectory through cumulative metrics

This complements the first figure which showed time-series learning curves, while this figure focuses on distributions and cumulative outcomes.

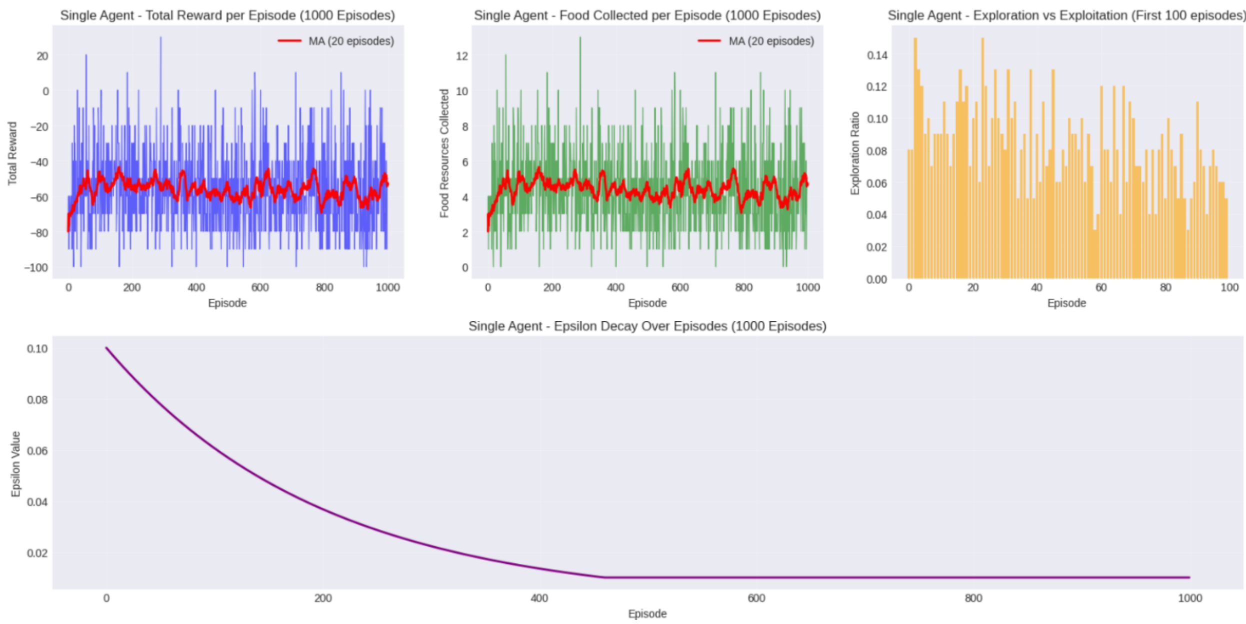

Figure3:Single Agent Q-Learning over 1000 Episodes

The figure contains these four panels:

1. **Top-left**: Total reward per episode over 1000 episodes with moving average
2. **Top-right**: Food resources collected per episode over 1000 episodes with moving average
3. **Bottom-left**: Exploration ratio (random vs greedy actions) for first 100 episodes
4. **Bottom-right**: Epsilon decay over the full 1000 episodes

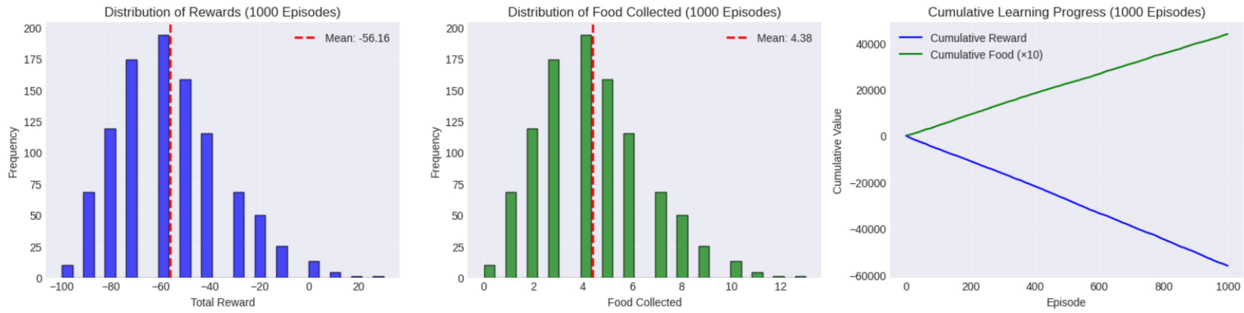

Figure4:Single Agent Performance Distributions and Cumulative Progress (1000 Episodes)

The figure contains these three panels:

1. **Left**: Histogram of reward distribution over 1000 episodes (mean: -56.16)
2. **Middle**: Histogram of food collected distribution over 1000 episodes (mean: 4.38)
3. **Right**: Cumulative learning progress showing both cumulative reward and cumulative food (scaled by 10)

4. Multi-Agent Learning Behavior

The multi-agent scenario includes **three agents** operating in the same grid environment.

4.1 Independent Q-Tables

Each agent maintains its own Q-table and learns independently.

Observations

- Agents compete for the same food resources.
- Frequent collisions occur, especially in early episodes.
- Learning is slower due to lack of shared experience.
- Reward variance is higher across episodes.

Interpretation

This setup demonstrates **competitive behavior**, as agents act selfishly without coordination.

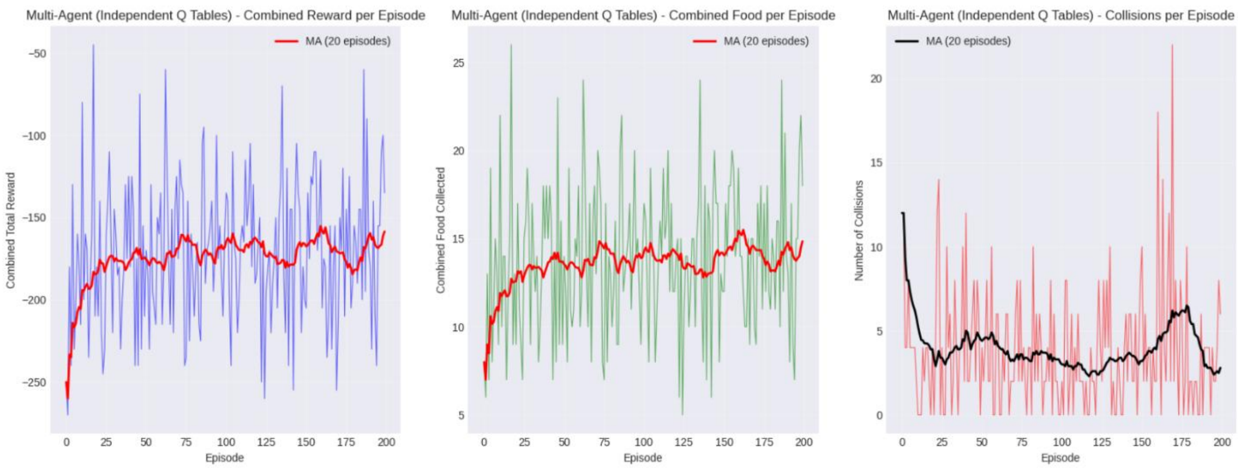

Figure5:Multi-Agent Learning with Independent Q-Tables

The figure contains these three panels:

1. **Left**: Combined total reward per episode for all 3 agents (showing negative values around -50 to -200)
2. **Middle**: Combined food collected per episode by all agents (around 5-10 food items)
3. **Right**: Number of collisions between agents per episode (varying from 0 to 15+)

Figure6:Multi-Agent Independent Learning: Individual Agent Performance Analysis

The figure contains these three panels:

1. **Left**: Food collected per agent in the last 50 episodes (showing bars for Agent 0, 1, 2 over recent episodes)
2. **Middle**: Bar chart comparing average performance of each agent (food vs scaled reward)
3. **Right**: Cumulative learning progress for the entire multi-agent system

4.2 Shared Q-Table

All agents share a single Q-table and update it collectively.

Observations

- Faster learning compared to independent Q-tables.
- Reduced collision rates over time.
- More efficient food collection.
- Smoother reward curves.

Interpretation

Sharing knowledge allows agents to benefit from each other's experiences, resulting in **implicit cooperation**, even without direct communication.

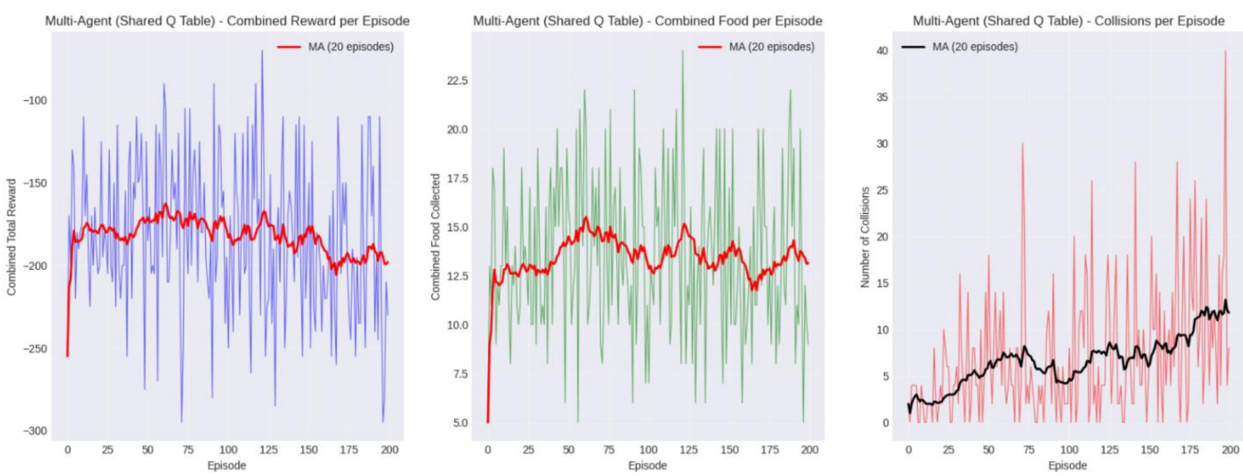

Figure7:Multi-Agent Learning with Shared Q-Table: Combined Performance (200 Episodes)

The figure contains these three panels:

1. **Left**: Combined total reward per episode for all 3 agents (very negative, around -250 to -300)
2. **Middle**: Combined food collected per episode (improving from ~10 to ~22.5 over time)
3. **Right**: Number of collisions between agents per episode (initially high ~40, decreasing to ~15)

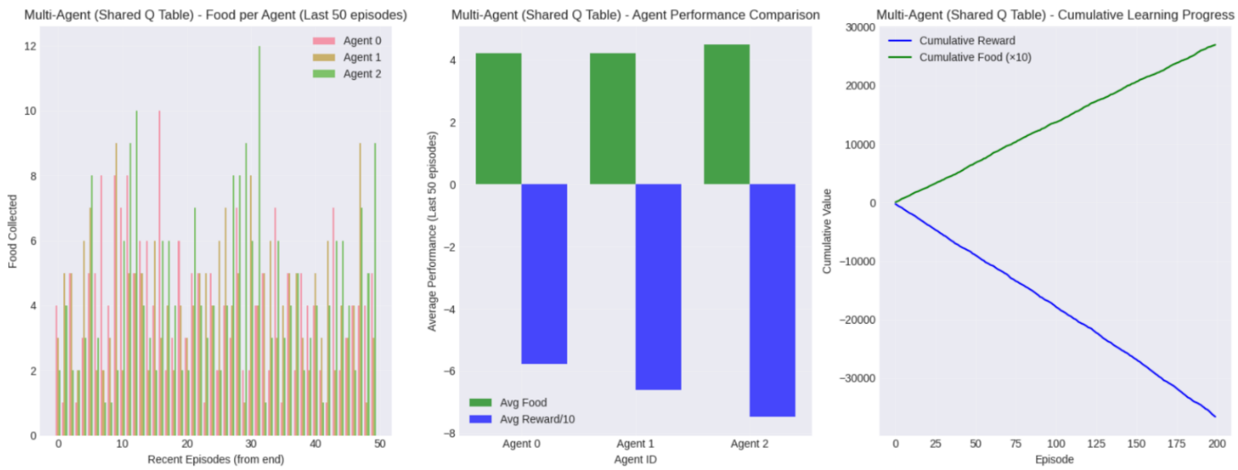

Figure8: Multi-Agent Learning with Shared Q-Table:Individual Performance and Collective Outcomes

The figure contains these three panels:

1. **Left**: Food collected per agent in the last 50 episodes (showing similar performance across agents)
2. **Middle**: Bar chart comparing average performance of each agent (showing very similar values)
3. **Right**: Cumulative learning progress for the entire system (showing negative cumulative reward)

4.3 Comparison: Independent vs. Shared Q-Tables

Aspect	Independent Q-Tables	Shared Q-Table
Learning speed	Slower	Faster
Collisions	Higher	Lower
Cooperation	None	Implicit
Stability	Less stable	More stable

Table1:Independent vs. Shared

The shared Q-table approach demonstrates better overall performance and coordination.

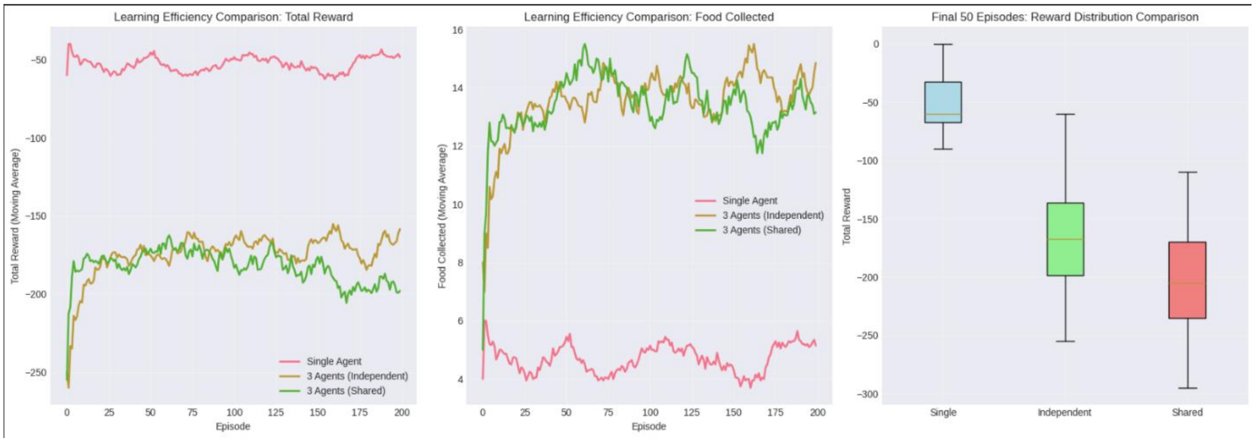

Figure9:Learning Efficiency Comparison: Single Agent vs. Multi-Agent Configurations

The figure contains these three panels:

1. **Left**: Learning efficiency comparison for total reward (moving averages of all three configurations)
2. **Middle**: Learning efficiency comparison for food collected (moving averages of all three configurations)
3. **Right**: Box plot comparison of reward distributions in final 50 episodes

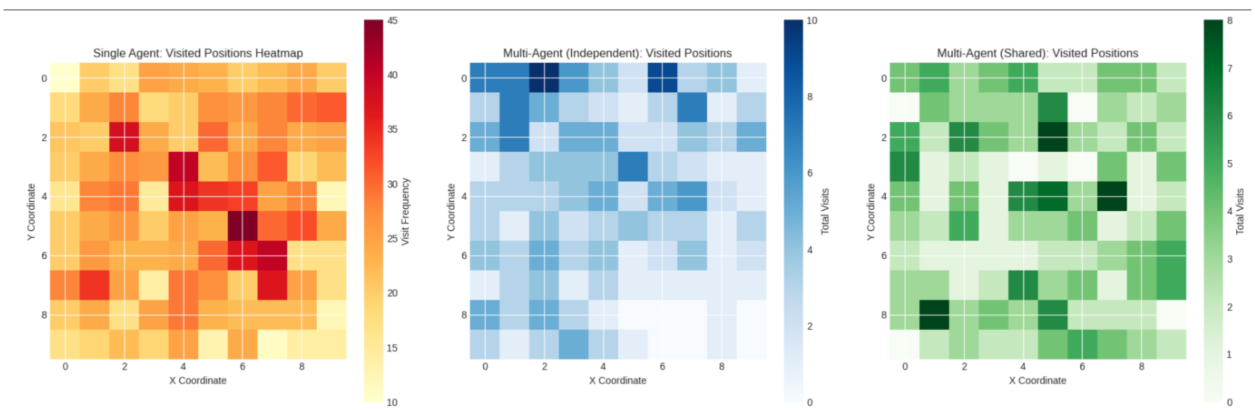

Figure10:Agent Spatial Exploration Patterns: Heatmap Comparison

The figure contains these three heatmap panels:

1. **Left**: Single agent visited positions heatmap
2. **Middle**: Multi-agent with independent Q-tables visited positions
3. **Right**: Multi-agent with shared Q-table visited positions

5. Conclusion and Future Improvements

Conclusion

This project demonstrated the effectiveness of Q-learning in both single-agent and multi-agent foraging environments. While single-agent learning converges efficiently, multi-agent systems introduce challenges such as competition and collisions. Sharing Q-tables significantly improves learning speed, stability, and cooperation.

Potential Improvements

- Introduce **communication between agents**.
- Use Deep Q-Networks **(DQN)** for larger state spaces.
- Implement **dynamic obstacles**.
- Increase the number of agents to study scalability.
- Explore **reward shaping** to further encourage cooperation.